

Frozen LLM → **residual stream at layer ℓ** : $x_{i,j} \in \mathbb{R}^d$, $j = 1 \dots N$, N up to 131,072



k -token attack needle inside benign context (dilution $k/N \approx 2.4 \times 10^{-4}$)

Softmax attention pooling

$$\alpha_j = \text{softmax}_j(q^\top y_j / \sqrt{m})$$

score tensor $(1, N)$ and $(1, N, m)$ features

materialised in full $\Rightarrow \Theta(N)$ memory

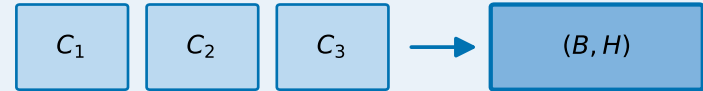
652.1 MiB overhead at $N = 131,072$

scaling exponent $\alpha = 0.842$

34 x

MultiMax aggregation (ours)

$$a_h = \max_j v_h^\top y_j, \quad h = 1 \dots H$$



streamed; only a running max is carried

19.1 MiB overhead at $N = 131,072$

$\Theta(\min(C, N))$, $\alpha = 0.000$

The cascade: MultiMax is the $\Theta(1)$ -memory first pass;
the expensive monitor sees only what the gate cannot resolve.

Platt-scaled cascading gate

$$\hat{p} = \sigma(az + b) \quad \text{escalate iff } |\hat{p} - 0.5| < \delta$$

δ set from an escalation budget, not hard-coded

confident → **probe verdict**

97.5% of traffic, $\Theta(1)$ memory in N

ambiguous → **LLM**

2.5%, full inspection